

TMFC050 Product Recommendation Management v1.1.0

On this page:

- [Notice](#)
- [1. Overview](#)
- [2. eTOM Processes, SID Data Entities and Functional Framework Functions](#)
 - [2.1. eTOM business activities](#)
 - [2.2. SID ABEs](#)
 - [2.3. eTOM L2 - SID ABEs links](#)
 - [2.4. Functional Framework Functions](#)
- [3. TM Forum Open APIs & Events](#)
 - [3.1. Exposed APIs](#)
 - [3.2. Dependent APIs](#)
 - [3.3. Events](#)
- [4. Machine Readable Component Specification](#)
- [5. References](#)
 - [5.1. TMF Standards related versions](#)
 - [5.2. Jira References](#)
 - [5.3. Further resources](#)
- [6. Administrative Appendix](#)
 - [6.1. Document History](#)
 - [6.1.1. Version History](#)
 - [6.1.2. Release History](#)
 - [6.2. Acknowledgements](#)

Maturity Level: General availability (GA)	Team Approved Date: 11-Mar-2025
Release Status: Pre-production	Approval Status: Team Approved
Version: 1.1.0	IPR Mode: RAND

Notice

Copyright © TM Forum 2025. All Rights Reserved.

This document and translations of it may be copied and furnished to others, and derivative works that comment on or otherwise explain it or assist in its implementation may be prepared, copied, published, and distributed, in whole or in part, without restriction of any kind, provided that the above copyright notice and this section are included on all such copies and derivative works. However, this document itself may not be modified in any way, including by removing the copyright notice or references to TM FORUM, except as needed for the purpose of developing any document or deliverable produced by a TM FORUM Collaboration Project Team (in which case the rules applicable to copyrights, as set forth in the [TM FORUM IPR Policy](#) must be followed) or as required to translate it into languages other than English.

The limited permissions granted above are perpetual and will not be revoked by TM FORUM or its successors or assigns.

This document and the information contained herein is provided on an "AS IS" basis and TM FORUM DISCLAIMS ALL WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTY THAT THE USE OF THE INFORMATION HEREIN WILL NOT INFRINGE ANY OWNERSHIP RIGHTS OR ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

Direct inquiries to the TM Forum office:

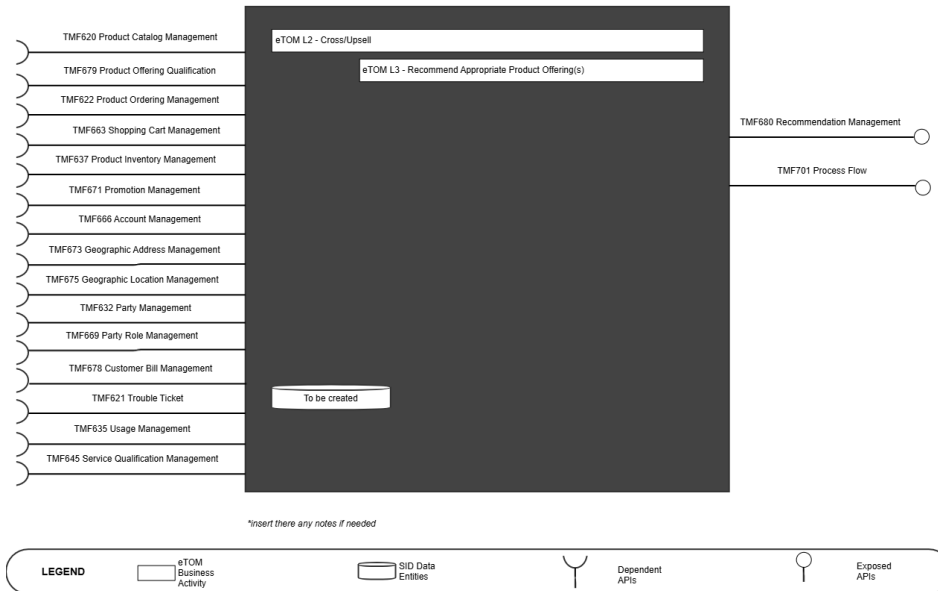
181 New Road, Suite 304
Parsippany, NJ 07054 USA
Tel No. +1 862 227 1648
TM Forum Web Page: www.tmforum.org

1. Overview

1. [TAC-208](#) IG1171 (update) Component Definition to v4.0.0 and incorporate IG1245 Principles to Define ODA Components
2. [\[TAC-250\] IG 1171 Improvements Some observations & recommendations.](#) - [TM Forum JIRA](#)
3. [\[TAC-214\] Interface Standardization needs all 3 stages of process to be developed](#) - [TM Forum JIRA](#)
4. [\[TAC-226\] Overview](#) - [TM Forum JIRA](#)
5. [ODA-846](#) Summary of ODA component Template enhancements for 14th Sep Review

Component Name	ID	Description	ODA Function Block
Product Recommendation Management	TMFC050	Product Recommendation component is responsible for evaluating and providing product offering/specification recommendations based on the history and real-time context of a party and marketing strategy.	Intelligence Management

TMFC050 : Product Recommendation Management



2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM business activities

eTOM business activities this ODA Component is responsible for.

Identifier	Level	Business Activity Name	Description
1.1.9.4	L2	Cross/Up Sell	Cross/Up Sell ensures that the value of the relationship between the customer or other party and an enterprise is maximized by selling additional, or more of the existing, product offerings. The ongoing analysis of customer or other party trends, such as usage, problems, complaints, is used to identify when the current offerings may no longer be appropriate or when the opportunity for a larger sale arises. Based on the data collected, more appropriate or other offering(s) may be recommended to the customer or other party.
1.1.9.4.2	L3	Recommend Appropriate Product Offering(s)	Recommend Appropriate Product Offering(s) recommends more appropriate or other product offering(s) to the customer or other party that may be based on the results of Analyze Customer or Other Party Trends or based on known other offerings that represent an upsell. For example, a potential new partner may be offered the opportunity to have there logo and contact information added to an enterprise's application that manages a IoT device.

2.2. SID ABEs

SID ABEs this ODA Component is responsible for:

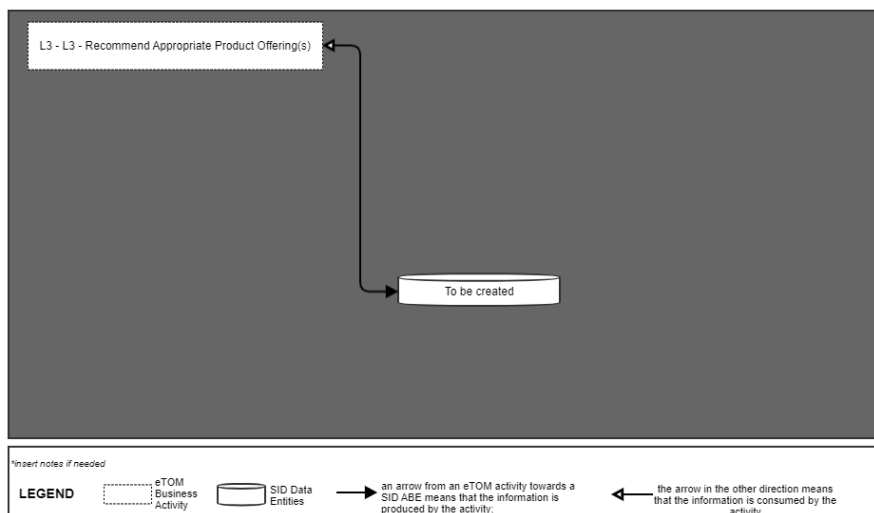
SID ABE Level 1	SID ABE Level 2 (or set of BEs)*

*: if SID ABE Level 2 is not specified this means that all the L2 business entities must be implemented, else the L2 SID ABE Level is specified.

Note: There is need to create Product Recommendation Management BE in SID.

2.3. eTOM L2 - SID ABEs links

eTOM L2 vS SID ABEs links for this ODA Component.



2.4. Functional Framework Functions

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
1196	Product Specification & Offering Simulation	Product Specification & Offering Simulation Function simulates Product Specification configuration and Product Offerings on the real traffic of the customer thanks to Tariff Calculation and Rating.	Product Operational Analysis	Pushed Offer Management
1195	Pushed Offer Identification	Pushed Offer Identification Function identifies offers adapted to the customer needs including personalized offers. Based on a customer's individualized knowledge (Marketing one to one), the aim is to elaborate for each customer according to his needs and recent action individualized Product Specifications & Offerings.	Product Operational Analysis	Pushed Offer Management

3. TM Forum Open APIs & Events

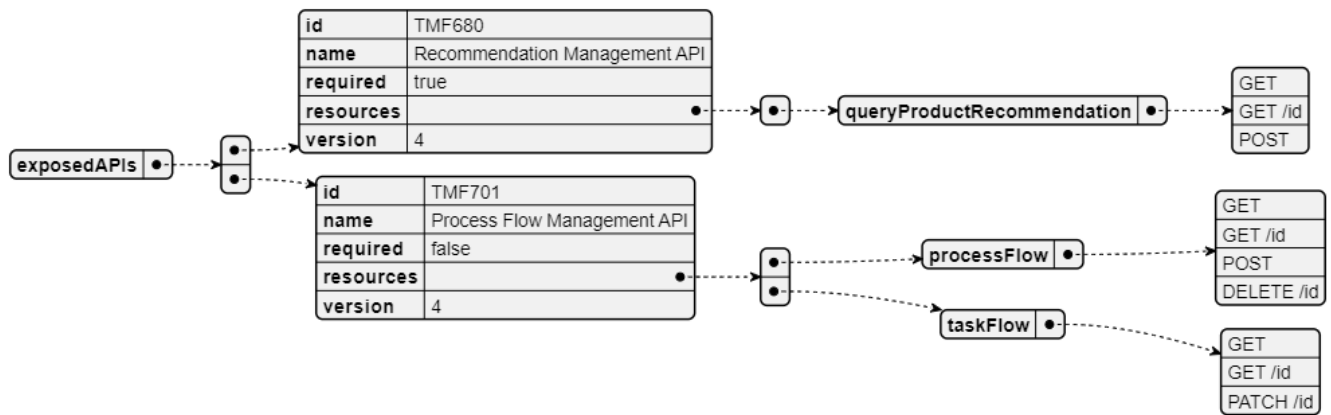
The following part covers the APIs and Events; This part is split in 3:

- List of **Exposed APIs** - This is the list of APIs available from this component.
- List of **Dependent APIs** - In order to satisfy the provided API, the component could require the usage of this set of required APIs.
- List of **Events (generated & consumed)** - The events which the component may generate is listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

<Note note to be inserted into ODA Component specifications: If a new Open API is required, but it does not yet exist. Then you should include a textual description of the new Open API, and it should be clearly noted that this Open API does not yet exist. In addition, a Jira epic should be raised to request the new Open API is added, and the Open API team should be consulted. Finally, a decision is required on the feasibility of the component without this Open API. If the Open API is critical then the component specification should not be published until the Open API issue has been resolved. Alternatively if the Open API is not critical, then the specification could continue to publication. The result of this decision should be clearly recorded.>

3.1. Exposed APIs

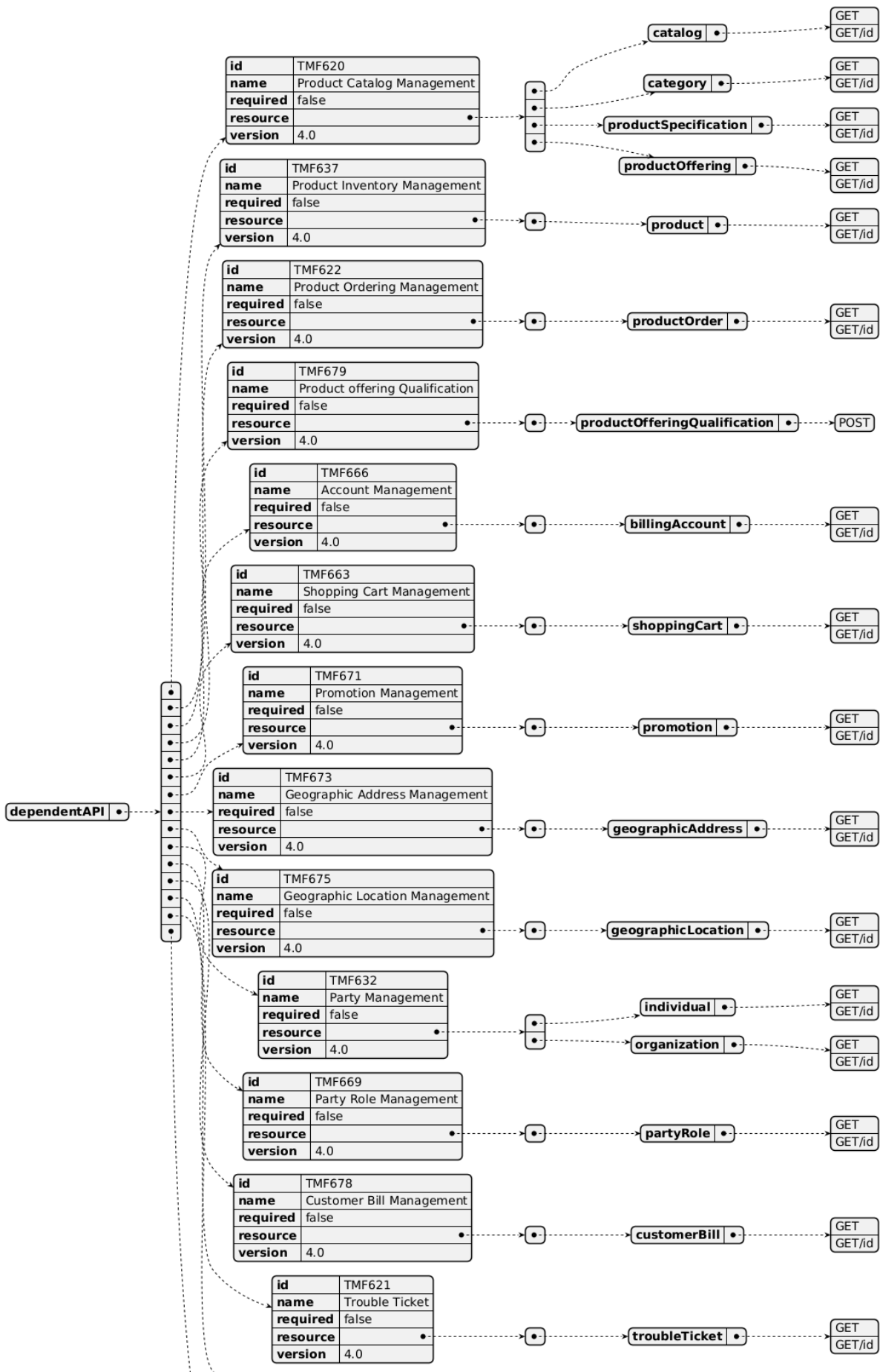
Following diagram illustrates API/Resource/Operation:

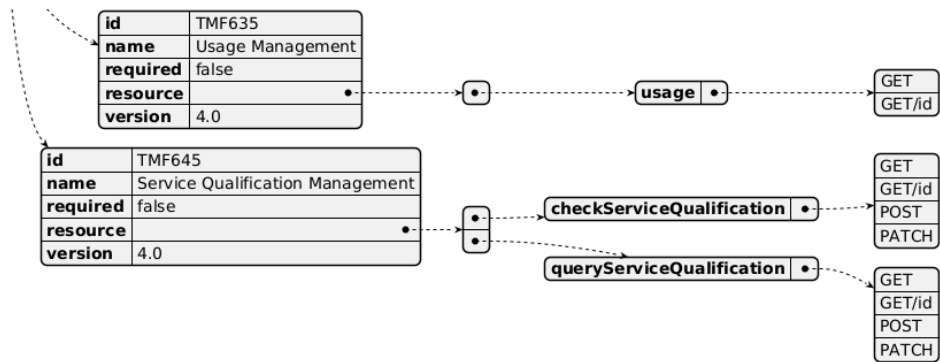


API ID	API Name	API Version	Mandatory / Optional	Operations
TMF680	Recommendation Management	4	Mandatory	GET GET /id POST
TMF701	Process Flow API	4	Optional	GET GET /id POST

3.2. Dependent APIs

Following diagram illustrates API/Resource/Operation potentially used by the product recommendation management component:



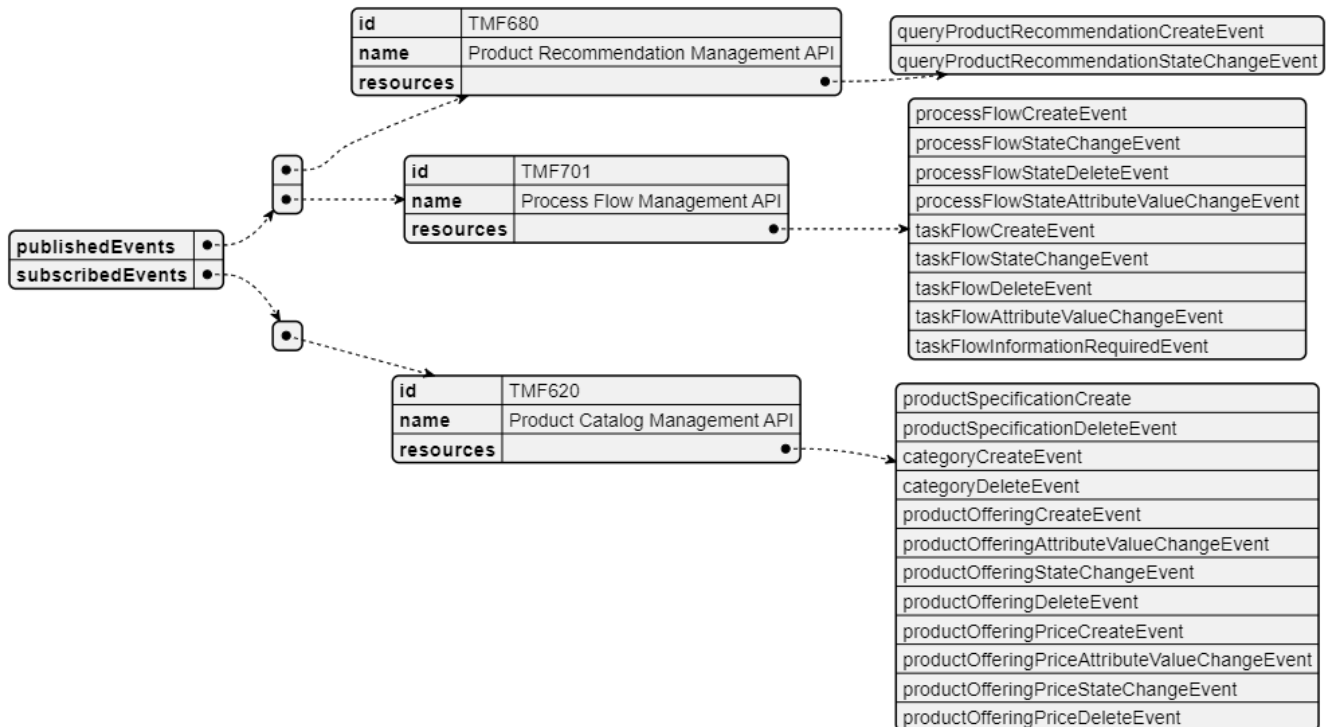


API ID	API Name	API Version	Mandatory / Optional	Operations	Rationales	
TMF620	Product Catalog Management	4	Mandatory	- catalog: - GET - GET /id - category: - GET - GET /id - productSpecification: - GET - GET /id - productOffering: - GET - GET /id - productOfferingPrice: - GET - GET /id		
TMF637	Product Inventory Management	4	Optional	- product: - GET - GET /id		
TMF622	Product Ordering Management	4	Optional	- productOrder: - GET - GET /id		
TMF679	Product Offering Qualification	4	Optional	- productOfferingQualification: - GET - GET /id		
TMF666	Account Management	4	Optional	- billingAccount: - GET - GET /id		
TMF663	Shopping Cart Management	4	Optional	- shoppingCart: - GET - GET /id		
TMF671	Promotion Management	4	Optional	- promotion: - GET - GET /id		
TMF673	Geographic Address Management	4	Optional	- geographicAddressManagement: - GET - GET /id		
TMF675	Geographic Location Management	4	Optional	- geographicLocationManagement: - GET - GET /id		
TMF632	Party Management	4	Optional	- individual: - GET - GET /id - organization: - GET - GET /id		
TMF669	Party Role Management	4	Optional	- partyRole: - GET - GET /id		
TMF678	Customer Bill Management	4	Optional	- customerBill: - GET - GET /id		
TMF621	Trouble Ticket	4	Optional	- troubleTicket: - GET - GET /id		
TMF635	Usage Management	4	Optional	- usage: - GET - GET /id		
TMF645	Service Qualification Management	4	Optional	checkServiceQualification: -Get -Get /id -Post -Patch		

				queryServiceQualification:		
				-Get		
				-Get /id		
				-Post		
				-Patch		

3.3. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.



4. Machine Readable Component Specification

Refer to the ODA Component table for the machine-readable component specification file for this component or click the link [Recommendation_Management V1.1.yaml](#).

5. References

5.1. TMF Standards related versions

Standard	Version(s)
SID	24.0
eTOM	24.0
Functional Framework	24.0

5.2. Jira References

- JIRA request for creating the SID entity for queryRecommendationManagement : [\[ISA-1194\] SID Entity for queryRecommendationManagement - TM Forum Jira](#)
- JIRA request for improving the function description of 1196: [\[FX-1227\] Improvement of the description for function id 1196 - Product Specification & Offering Simulation - TM Forum Jira](#)
- JIRA request for improving the function description of 1195: [\[FX-1228\] Improvement of the description for function id 1195 - Pushed Offer Identification - TM Forum JIRA](#)

5.3. Further resources

1. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
1.0.0	11 Dec 2024	Abinash Vishwakarma	First version.
1.0.0	07 Jan 2025	Rosie Wilson	Final administrative edits prior to publication
1.1.0	10 Mar 2025	Abinash Vishwakarma	Added Service Qualification Management
1.1.0	14 Mar 2025	Rosie Wilson	Final administrative edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	07 Jan 2025	Rosie Wilson	Initial Release
Production	07-Mar-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	14 Mar 2025	Rosie Wilson	New version 1.1.0

6.2. Acknowledgements

This document was prepared by the members of the TM Forum Component and Canvas project team:

Member	Company	Role*
Abinash Vishwakarma	Netcracker	Editor
Gaetano Biancardi	Accenture	Reviewer
Sylvie Demarest	Orange	Reviewer

*Select from: Project Chair, Project Co-Chair, Author, Editor, Key Contributor, Additional Input, Reviewer

© TM Forum 2025. All Rights Reserved.